

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Specimen 4WM2H Worked Solutions

Specimen | 4WM2H | Unit 2

29 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3**TOPIC: **Statistics Toolkit**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 1** Here are five cards, where x represents a number.

15	7	-2	23	x
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The mean of the five numbers is 12

Work out the value of x

$x =$

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION**Q#: 1****Marks: 3****TOPIC: Statistics Toolkit**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate statistic**

The mean is 12, so the total of the five numbers is

$$5 \times 12 = 60$$

2. Calculate value

The known numbers add to

$$15 + 7 - 2 + 23 = 43$$

3. Calculate value

Therefore

$$43 + x = 60$$

$$x = 17$$

FINAL ANSWER

$$x = 17.$$

PAST PAPER QUESTION**Q#: 2****Marks: 4****TOPIC: Prime Factors, HCF & LCM**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 2 (a) Find the highest common factor (HCF) of 56 and 84
Show your working clearly.

.....
(2)

- (b) Find the lowest common multiple (LCM) of 60 and 72
Show your working clearly.

.....
(2)

.....
(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 4****TOPIC: Prime Factors, HCF & LCM**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the LCM**

$$56 = 2^3 \times 7, \quad 84 = 2^2 \times 3 \times 7$$

2. The common prime factors areThe common prime factors are 2^2 and 7, so

$$\text{HCF} = 2^2 \times 7 = 28$$

3. Find the LCM

Also

$$60 = 2^2 \times 3 \times 5, \quad 72 = 2^3 \times 3^2$$

4. Find the LCM

Use the highest powers of each prime:

$$\text{LCM} = 2^3 \times 3^2 \times 5 = 360$$

FINAL ANSWER

HCF = 28, LCM = 360.

PAST PAPER QUESTION**Q#: 3****Marks: 2****TOPIC: Algebraic Roots & Indices**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

3 $\frac{2^x}{4^n} = 2^x$

Find an expression for x in terms of k and n

$x = \dots\dots\dots$

(Total for Question 3 is 2 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 2**TOPIC: **Algebraic Roots & Indices**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use index laws**

$$4^n = (2^2)^n = 2^{2n}$$

2. Use index laws

So

$$\frac{2^k}{4^n} = \frac{2^k}{2^{2n}} = 2^{k-2n}$$

3. Use index lawsSince this equals 2^x ,

$$x = k - 2n$$

FINAL ANSWER

$$x = k - 2n.$$

PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 4 The diagram shows parts of three regular polygons, **A**, **B** and **C**, meeting at a point.

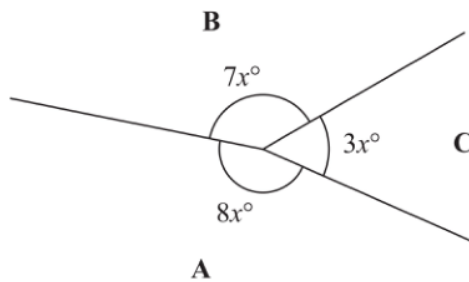


Diagram **NOT**
accurately drawn

Polygon **B** has n sides.
Work out the value of n

$n =$

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

1. Angles around a point addAngles around a point add to 360° :

$$8x + 7x + 3x = 360$$

$$18x = 360$$

$$x = 20$$

2. The interior angle of polygonThe interior angle of polygon B is

$$7x = 7(20) = 140^\circ$$

3. Find the gradientFor a regular n -sided polygon,

$$\frac{180(n-2)}{n} = 140$$

$$180n - 360 = 140n$$

$$40n = 360$$

$$n = 9$$

FINAL ANSWER

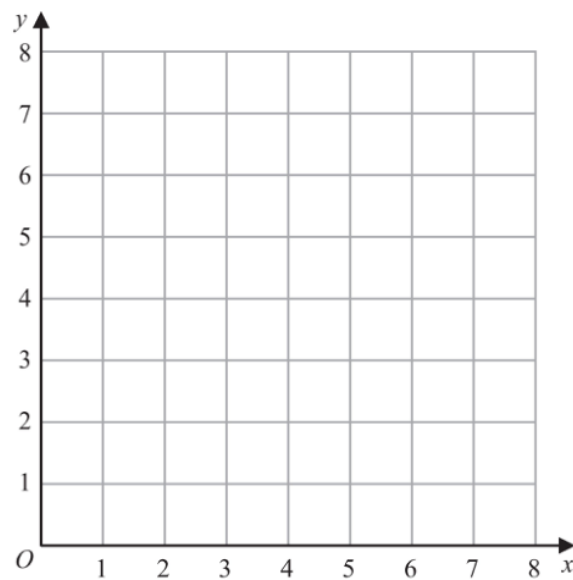
$$n = 9.$$

PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC: Graphing Inequalities**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

5 (a) On the grid, draw and label with its equation the straight line with equation

(i) $y = 1$ (ii) $x = 2$ (iii) $x + y = 7$



(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$y \geq 1$ $x \geq 2$ $x + y \leq 7$

Label the region **R**

(1)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 4****TOPIC: Graphing Inequalities**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Draw the three boundary lines**

Draw the three boundary lines:

$$y = 1, \quad x = 2, \quad x + y = 7$$

2. Read the graphFor $x + y = 7$, use points such as $(0, 7)$ and $(7, 0)$.**3. The required region is**

The required region is:

$$y \geq 1, \quad x \geq 2, \quad x + y \leq 7$$

4. Read the graphSo shade the triangular region above $y = 1$, to the right of $x = 2$, and below the line $x + y = 7$. Its vertices are

$$(2, 1), \quad (2, 5), \quad (6, 1)$$

FINAL ANSWERthe shaded region R is the triangle with vertices $(2, 1)$, $(2, 5)$, and $(6, 1)$.

PAST PAPER QUESTION**Q#: 6****Marks: 3**TOPIC: **Statistics Toolkit**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

6 Here are some integers where $a < b < c < d$

a b c d d d

The mode of the integers is 9

The range of the integers is 4

The median of the integers is 8

Work out the value of a , the value of b , the value of c and the value of d

$a =$

$b =$

$c =$

$d =$

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 3****TOPIC: Statistics Toolkit**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate statistic**

The mode is 9, and d appears three times, so

$$d = 9$$

2. Calculate statistic

The range is 4, so

$$d - a = 4$$

$$9 - a = 4$$

$$a = 5$$

3. Calculate statistic

There are six values, so the median is the mean of the 3rd and 4th values:

$$\frac{c + d}{2} = 8$$

$$\frac{c + 9}{2} = 8$$

$$c = 7$$

4. Solve inequality

Since $a < b < c < d$, the remaining integer is $b = 6$.

FINAL ANSWER

$a = 5, b = 6, c = 7, d = 9.$

PAST PAPER QUESTION**Q#: 7** **Marks: 4****TOPIC: Standard & Compound Units**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 7 A cylinder is placed on the ground.

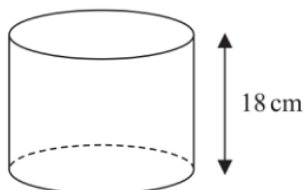


Diagram **NOT**
accurately drawn

The height of the cylinder is 18 cm

The force exerted by the cylinder on the ground is 72 newtons.

The pressure on the ground due to the cylinder is 1.4 newtons/cm²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cylinder.

Give your answer correct to 3 significant figures.

..... cm³

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 4****TOPIC: Standard & Compound Units**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate area**

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

2. Calculate area

So the area of the base is

$$\text{area} = \frac{72}{1.4} = 51.428571 \dots$$

3. The volume is

The volume is

$$51.428571 \dots \times 18 = 925.714285 \dots$$

4. Calculate volume

Correct to 3 significant figures,

$$925.714 \dots \approx 926$$

FINAL ANSWER926 cm³.

PAST PAPER QUESTION**Q#: 8****Marks: 6****TOPIC: Compound Interest & Depreciation**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 8** In 2021, the value of Asha's apartment was 634 400 euros.
The value of Asha's apartment had increased by 4% from its value in 2020

(a) Work out the value of Asha's apartment in 2020

..... euros
(3)

Pam bought a boat.

In each year after Pam bought the boat, the value of the boat depreciated by 15%

- (b) Work out the total percentage by which the value of the boat had depreciated by the end of the second year after Pam bought the boat.

..... %
(3)

(Total for Question 8 is 6 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 6****TOPIC: Compound Interest & Depreciation**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use compound interest**

For part (a), the 2021 value is 104% of the 2020 value.

$$\text{2020 value} = \frac{634400}{1.04} = 610000$$

2. Use compound interest

For part (b), after one year the multiplier is 0.85. After two years:

$$0.85^2 = 0.7225$$

3. Use compound interest

So 72.25% of the value remains, and the total depreciation is

$$100\% - 72.25\% = 27.75\%$$

FINAL ANSWER

610000 euros; 27.75%.

PAST PAPER QUESTION**Q#: 9****Marks: 2****TOPIC: Powers, Roots & Standard Form**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

9 (a) Write 0.000089 in standard form.

.....
(1)

(b) Write 8.34×10^4 as an ordinary number.

.....
(1)

(Total for Question 9 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 9

Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert standard form**

$$0.000089 = 8.9 \times 10^{-5}$$

2. Convert standard form

Also

$$8.34 \times 10^4 = 83400$$

FINAL ANSWER 8.9×10^{-5} , 83400.

EA

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Ratio Problem Solving**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

10 Payel makes 300 celebration cards so that

number of number of number of
 birthday cards : anniversary cards : congratulations cards = 7 : 5 : 3

$\frac{2}{5}$ of the birthday cards have numbers on them.

36% of the anniversary cards have numbers on them.

None of the congratulations cards have numbers on them.

Work out what fraction of the 300 cards have numbers on them.
Give your answer in its simplest form.

.....
(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION**Q#: 10** **Marks: 5****TOPIC: Ratio Problem Solving**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. The total number of parts**

The total number of parts is

$$7 + 5 + 3 = 15$$

2. Each part is

Each part is

$$300 \div 15 = 20$$

3. Split the ratio

So the numbers of cards are:

$$140, \quad 100, \quad 60$$

4. The birthday cards with numbers

The birthday cards with numbers are

$$\frac{2}{5} \times 140 = 56$$

5. The anniversary cards with numbers

The anniversary cards with numbers are

$$36\% \times 100 = 36$$

6. Split the ratio

So the total with numbers is

$$56 + 36 = 92$$

7. The fraction of all cards

The fraction of all cards is

$$\frac{92}{300} = \frac{23}{75}$$

FINAL ANSWER

$$\frac{23}{75}$$

PAST PAPER QUESTION**Q#: 11****Marks: 3****TOPIC: Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

11 Solve the simultaneous equations

$$7x + 3y = 3$$

$$3x - y = 7$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 3****TOPIC: Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve simultaneous equations**

From

$$3x - y = 7$$

$$y = 3x - 7$$

2. Solve simultaneous equationsSubstitute into $7x + 3y = 3$:

$$7x + 3(3x - 7) = 3$$

$$16x - 21 = 3$$

$$16x = 24$$

$$x = 1.5$$

3. Solve simultaneous equations

Then

$$y = 3(1.5) - 7 = -2.5$$

FINAL ANSWER

$$x = 1.5, y = -2.5.$$

PAST PAPER QUESTION**Q#: 12****Marks: 5****TOPIC: Percentages**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

12 Zimo is going on holiday.

He makes 3 separate payments to cover the total cost of his holiday.

The following table shows how much money Zimo pays to the holiday company.

Payment	Amount paid
Payment 1	$\frac{2}{5}$ of the total cost
Payment 2	45% of the total cost
Payment 3	\$405

Work out how much Zimo has to pay for Payment 2

\$.....

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 5****TOPIC: Percentages**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate percentage**

$$\frac{2}{5} = 40\%$$

2. Calculate percentage

So Payments 1 and 2 together are

$$40\% + 45\% = 85\%$$

3. This means Payment 3 is

This means Payment 3 is

$$100\% - 85\% = 15\%$$

4. Calculate percentage

So

$$15\% = 405$$

$$1\% = 27$$

5. Payment 2 is

Payment 2 is 45%:

$$45 \times 27 = 1215$$

FINAL ANSWER

\$1215.

PAST PAPER QUESTION**Q#: 13****Marks: 4****TOPIC: Functions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

13 The function f is defined as

$$f : x \mapsto \frac{2x}{x-6}$$

(a) Find $f(10)$
(1)(b) Express the inverse function f^{-1} in the form $f^{-1} : x \mapsto \dots$

$$f^{-1} : x \mapsto \dots$$

(3)

PAST PAPER SOLUTION**Q#: 13****Marks: 4****TOPIC: Functions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$f(10) = \frac{2(10)}{10 - 6} = \frac{20}{4} = 5$$

2. Let

For the inverse, let

$$y = \frac{2x}{x - 6}$$

$$y(x - 6) = 2x$$

$$xy - 6y = 2x$$

$$x(y - 2) = 6y$$

$$x = \frac{6y}{y - 2}$$

3. Evaluate fraction

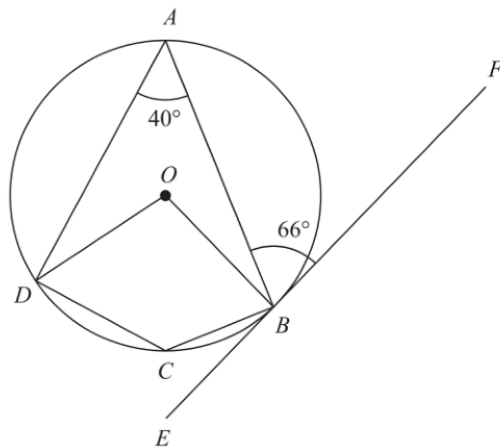
So

$$f^{-1} : x \mapsto \frac{6x}{x - 2}$$

FINAL ANSWER5, and $f^{-1} : x \mapsto \frac{6x}{x-2}$.

PAST PAPER QUESTION**Q#: 14****Marks: 5****TOPIC: Circle Theorems**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

14 A, B, C and D are points on a circle, centre O EBF is the tangent to the circle at B Diagram **NOT**
accurately drawn(a) (i) Work out the size of angle DCB
(1)

(ii) Give a reason for your answer to (a)(i)

.....
(1)(b) Work out the size of angle ADO
(3)**(Total for Question 14 is 5 marks)**

PAST PAPER SOLUTION**Q#: 14****Marks: 5****TOPIC: Circle Theorems**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**In cyclic quadrilateral $ADCB$, opposite angles add to 180° .

$$\angle DCB = 180^\circ - 40^\circ = 140^\circ$$

2. Reason opposite angles in aReason: opposite angles in a cyclic quadrilateral add to 180° .**3. Use trigonometry**

By the alternate segment theorem,

$$\angle ADB = 66^\circ$$

4. Use trigonometryIn triangle ADB ,

$$\angle ABD = 180^\circ - 40^\circ - 66^\circ = 74^\circ$$

5. The angle at the centreThe angle at the centre is twice the angle at the circumference standing on the same chord AD :

$$\angle AOD = 2(74^\circ) = 148^\circ$$

6. Use trigonometrySince $OA = OD$, triangle AOD is isosceles:

$$\angle ADO = \frac{180^\circ - 148^\circ}{2} = 16^\circ$$

FINAL ANSWER $\angle DCB = 140^\circ$; reason: opposite angles in a cyclic quadrilateral; $\angle ADO = 16^\circ$.

PAST PAPER QUESTION

Q#: 15 Marks: 7

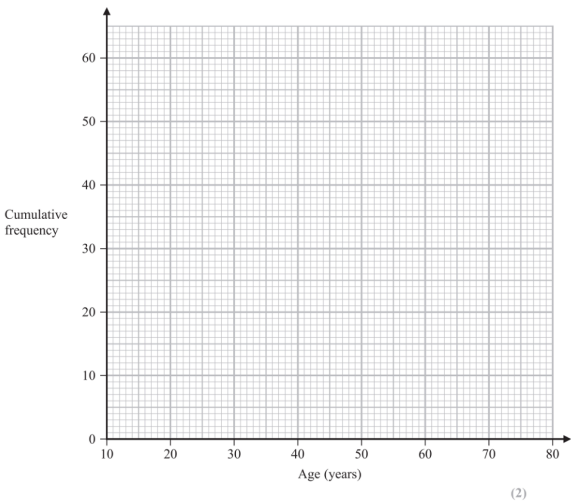
TOPIC: Cumulative Frequency Diagrams

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

15 The cumulative frequency table shows information about the ages of 60 people who went to a gym on Saturday.

Age (a years)	Cumulative frequency
$10 < a \leq 20$	13
$10 < a \leq 30$	36
$10 < a \leq 40$	42
$10 < a \leq 50$	47
$10 < a \leq 60$	52
$10 < a \leq 70$	56
$10 < a \leq 80$	60

(a) On the grid, draw a cumulative frequency graph for the information in the table.



(b) Use your graph to find an estimate for the median of the ages of these people.

_____ years
(1)

(c) Use your graph to find an estimate for the interquartile range of the ages of these people.

_____ years
(2)

(d) Use your graph to find an estimate for the number of these people who are older than 55 years.

_____ (2)

(Total for Question 15 is 7 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 7****TOPIC: Cumulative Frequency Diagrams**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Plot the cumulative frequency points**

Plot the cumulative frequency points and join with a smooth increasing curve:

$$(10, 0), (20, 13), (30, 36), (40, 42), (50, 47), (60, 52), (70, 56), (80, 60)$$

2. Use cumulative frequency

There are 60 people, so the median is read at cumulative frequency 30.

3. Convert standard form

Using linear interpolation between (20, 13) and (30, 36):

$$20 + \frac{30 - 13}{36 - 13} \times 10 \approx 27.4$$

4. Use cumulative frequency

So the median is about 27 years.

5. Use cumulative frequencyFor the interquartile range, read Q_1 at cumulative frequency 15 and Q_3 at cumulative frequency 45:

$$Q_1 \approx 20.9, \quad Q_3 \approx 46$$

$$\text{IQR} \approx 46 - 20.9 = 25.1$$

6. Use cumulative frequency

So the IQR is about 25 years.

7. Use cumulative frequency

At age 55, the cumulative frequency is about 49.5, so the number older than 55 is

$$60 - 49.5 \approx 10.5$$

8. This gives about people

This gives about 11 people.

FINAL ANSWER

median about 27 years, IQR about 25 years, about 11 people older than 55.

PAST PAPER QUESTION**Q#: 16****Marks: 4****TOPIC: Direct & Inverse Proportion**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

16 M is directly proportional to h^3
 $M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

$h = \dots\dots\dots$

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 4****TOPIC: Direct & Inverse Proportion**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find inverse function**Since M is directly proportional to h^3 ,

$$M = kh^3$$

2. Use whenUse $M = 4$ when $h = 0.5$:

$$4 = k(0.5)^3$$

$$4 = 0.125k$$

$$k = 32$$

3. UseNow use $M = 500$:

$$500 = 32h^3$$

$$h^3 = 15.625$$

$$h = 2.5$$

FINAL ANSWER

$$h = 2.5.$$

PAST PAPER QUESTION**Q#: 17****Marks: 5****TOPIC: Differentiation**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 17** A particle P moves along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds, $t \geq 1$, is s metres where

$$s = 4t^2 + \frac{125}{t}$$

The velocity of P at time t seconds, $t \geq 1$, is v m/s

Work out the distance of P from O at the instant when $v = 0$

..... m

(Total for Question 17 is 5 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 5****TOPIC: Differentiation**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Differentiate**

$$s = 4t^2 + \frac{125}{t}$$

2. Differentiate to find velocity

Differentiate to find velocity:

$$v = \frac{ds}{dt} = 8t - \frac{125}{t^2}$$

3. SetSet $v = 0$:

$$8t - \frac{125}{t^2} = 0$$

$$8t^3 = 125$$

$$t^3 = \frac{125}{8}$$

$$t = 2.5$$

4. DifferentiateNow find s :

$$s = 4(2.5)^2 + \frac{125}{2.5}$$

$$s = 25 + 50 = 75$$

FINAL ANSWER

75 m.

PAST PAPER QUESTION**Q#: 18****Marks: 3**TOPIC: **Solving Inequalities**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 18** Solve the inequality $2y^2 - 7y - 30 \leq 0$
Show your working clearly.

.....
(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 3****TOPIC: Solving Inequalities**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Factorise**

Factorise:

$$2y^2 - 7y - 30 = (2y + 5)(y - 6)$$

2. Solve inequality

So

$$(2y + 5)(y - 6) \leq 0$$

3. The roots are

The roots are

$$y = -\frac{5}{2}, \quad y = 6$$

4. Solve quadratic equation

The quadratic opens upwards, so it is less than or equal to zero between the roots.

FINAL ANSWER

$$-\frac{5}{2} \leq y \leq 6.$$

PAST PAPER QUESTION**Q#: 19****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

19 The diagram shows two similar metal statues.**A****B**Diagram **NOT**
accurately drawn

The volume of statue **B** is 20% less than the volume of statue **A**
 The surface area of statue **B** is $k\%$ less than the surface area of statue **A**

Work out the value of k
 Give your answer correct to 3 significant figures.

 $k = \dots\dots\dots$ **(Total for Question 19 is 4 marks)**

PAST PAPER SOLUTION**Q#: 19****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. The volume of statue is**

The volume of statue B is 80% of the volume of statue A , so the volume scale factor is

$$0.8$$

2. Calculate area

For similar solids:

$$\text{linear scale factor} = \sqrt[3]{0.8}$$

3. The surface area scale factor

The surface area scale factor is therefore

$$\left(\sqrt[3]{0.8}\right)^2 = 0.8^{2/3}$$

$$0.8^{2/3} = 0.861773\dots$$

4. Convert standard form

So the percentage decrease in surface area is

$$(1 - 0.861773\dots) \times 100 = 13.8226\dots$$

5. Calculate area

Correct to 3 significant figures,

$$k = 13.8$$

FINAL ANSWER

$$k = 13.8.$$

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

20 Solve the simultaneous equations

$$x - 2y = 3$$

$$x^2 - y^2 + 2x = 10$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 5****TOPIC: Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve simultaneous equations**

From

$$x - 2y = 3$$

$$x = 2y + 3$$

2. Solve simultaneous equationsSubstitute into $x^2 - y^2 + 2x = 10$:

$$(2y + 3)^2 - y^2 + 2(2y + 3) = 10$$

$$4y^2 + 12y + 9 - y^2 + 4y + 6 = 10$$

$$3y^2 + 16y + 5 = 0$$

$$(3y + 1)(y + 5) = 0$$

3. Solve simultaneous equations

So

$$y = -\frac{1}{3} \quad \text{or} \quad y = -5$$

4. Solve simultaneous equationsIf $y = -\frac{1}{3}$,

$$x = 2\left(-\frac{1}{3}\right) + 3 = \frac{7}{3}$$

5. Solve simultaneous equationsIf $y = -5$,

$$x = 2(-5) + 3 = -7$$

FINAL ANSWER $\left(\frac{7}{3}, -\frac{1}{3}\right)$ and $(-7, -5)$.

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

20 Solve the simultaneous equations

$$x - 2y = 3$$

$$x^2 - y^2 + 2x = 10$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 5****TOPIC: Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve simultaneous equations**

From

$$x - 2y = 3$$

$$x = 2y + 3$$

2. Solve simultaneous equationsSubstitute into $x^2 - y^2 + 2x = 10$:

$$(2y + 3)^2 - y^2 + 2(2y + 3) = 10$$

$$4y^2 + 12y + 9 - y^2 + 4y + 6 = 10$$

$$3y^2 + 16y + 5 = 0$$

$$(3y + 1)(y + 5) = 0$$

3. Solve simultaneous equations

So

$$y = -\frac{1}{3} \quad \text{or} \quad y = -5$$

4. Solve simultaneous equationsIf $y = -\frac{1}{3}$,

$$x = 2\left(-\frac{1}{3}\right) + 3 = \frac{7}{3}$$

5. Solve simultaneous equationsIf $y = -5$,

$$x = 2(-5) + 3 = -7$$

FINAL ANSWER $\left(\frac{7}{3}, -\frac{1}{3}\right)$ and $(-7, -5)$.

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Algebraic Fractions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

$$21 \quad a = \frac{14}{3x-7} \quad x = \frac{1}{4y-3}$$

Express a in the form $\frac{py+q}{ry+s}$ where p , q , r and s are integers.

Give your answer in its simplest form.

 $a = \dots\dots\dots$

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 3****TOPIC: Algebraic Fractions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify fraction**

$$a = \frac{14}{3x - 7}, \quad x = \frac{7}{4y - 3}$$

2. Simplify fractionSubstitute for x :

$$a = \frac{14}{3\left(\frac{7}{4y-3}\right) - 7}$$

3. Simplify the denominator

Simplify the denominator:

$$\begin{aligned} 3\left(\frac{7}{4y-3}\right) - 7 &= \frac{21}{4y-3} - 7 \\ &= \frac{21 - 7(4y-3)}{4y-3} = \frac{42 - 28y}{4y-3} \end{aligned}$$

4. Simplify fraction

So

$$\begin{aligned} a &= \frac{14}{\frac{42-28y}{4y-3}} = \frac{14(4y-3)}{42-28y} \\ a &= \frac{4y-3}{3-2y} \end{aligned}$$

FINAL ANSWER

$$a = \frac{4y-3}{3-2y}.$$

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Algebraic Fractions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

$$21 \quad a = \frac{14}{3x-7} \quad x = \frac{1}{4y-3}$$

Express a in the form $\frac{py+q}{ry+s}$ where p , q , r and s are integers.

Give your answer in its simplest form.

 $a = \dots\dots\dots$

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 3****TOPIC: Algebraic Fractions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify fraction**

$$a = \frac{14}{3x - 7}, \quad x = \frac{7}{4y - 3}$$

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4. Simplify fraction

So

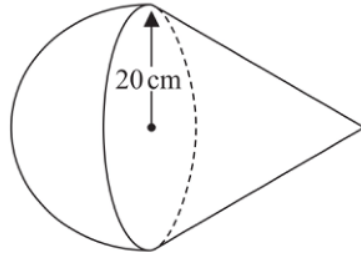
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FINAL ANSWER

$$a = \frac{4y-3}{3-2y}.$$

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Volume & Surface Area**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

22 A solid is made from a cone and a hemisphere.Diagram **NOT**
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm

The curved surface area of the cone is $580\pi \text{ cm}^2$

The volume of the solid is $k\pi \text{ cm}^3$

Work out the exact value of k

 $k = \dots\dots\dots$ **(Total for Question 22 is 5 marks)**

PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Volume & Surface Area**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate surface area

The curved surface area of a cone is

$$\pi r l$$

2. Calculate areaHere $r = 20$ and $\pi r l = 580\pi$, so

$$20l = 580$$

$$l = 29$$

3. Use Pythagoras to find the

Use Pythagoras to find the height of the cone:

$$h^2 = 29^2 - 20^2$$

$$h^2 = 841 - 400 = 441$$

$$h = 21$$

4. Volume of the cone

Volume of the cone:

$$\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(20)^2(21) = 2800\pi$$

5. Volume of the hemisphere

Volume of the hemisphere:

$$\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(20)^3 = \frac{16000}{3}\pi$$

6. Total volume

Total volume:

$$2800\pi + \frac{16000}{3}\pi = \frac{24400}{3}\pi$$

7. Calculate areaSince the volume is $k\pi$,

$$k = \frac{24400}{3}$$

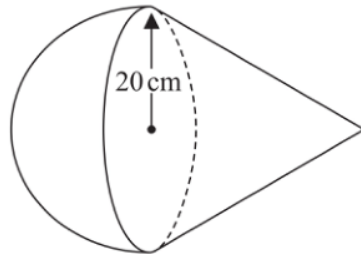
FINAL ANSWER

$$k = \frac{24400}{3}.$$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Volume & Surface Area**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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The curved surface area of the cone is $580\pi \text{ cm}^2$

The volume of the solid is $k\pi \text{ cm}^3$

Work out the exact value of k

 $k = \dots\dots\dots$ **(Total for Question 22 is 5 marks)**

PAST PAPER SOLUTION**Q#: 22****Marks: 5****TOPIC: Volume & Surface Area**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate surface area**

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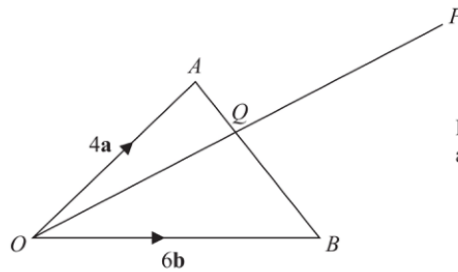
FINAL ANSWER

$$k = \frac{24400}{3}.$$

EA

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Vectors**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

23Diagram **NOT**
accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ : QB$

$$AQ : QB = \dots\dots\dots$$

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Vectors**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use vectors

$$\overrightarrow{OA} = 4a, \quad \overrightarrow{OB} = 6b, \quad \overrightarrow{AP} = 2a + 8b$$

2. Use vectors

So

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$$

$$\overrightarrow{OP} = 4a + (2a + 8b) = 6a + 8b$$

3. Use vectorsSince O, Q, P are collinear, let

$$\overrightarrow{OQ} = \lambda(6a + 8b)$$

4. Use vectorsAlso Q lies on AB . Let

$$\overrightarrow{AQ} = \mu\overrightarrow{AB}$$

5. Use vectors

Then

$$\overrightarrow{AB} = 6b - 4a$$

$$\overrightarrow{OQ} = 4a + \mu(6b - 4a)$$

$$\overrightarrow{OQ} = (4 - 4\mu)a + 6\mu b$$

6. Use vectors

Compare coefficients with

$$\lambda(6a + 8b) = 6\lambda a + 8\lambda b$$

7. Use vectors

So

$$4 - 4\mu = 6\lambda, \quad 6\mu = 8\lambda$$

8. Use vectors

From $6\mu = 8\lambda$,

$$\lambda = \frac{3}{4}\mu$$

9. Substitute

Substitute:

$$4 - 4\mu = 6\left(\frac{3}{4}\mu\right)$$

$$4 - 4\mu = \frac{9}{2}\mu$$

$$8 = 17\mu$$

$$\mu = \frac{8}{17}$$

10. Use vectors

Therefore

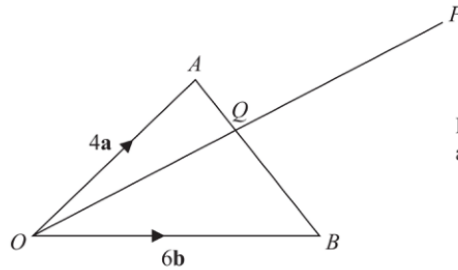
$$AQ : QB = \frac{8}{17} : \frac{9}{17} = 8 : 9$$

FINAL ANSWER

$$AQ : QB = 8 : 9.$$

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Vectors**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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$$\vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ : QB$

$$AQ : QB = \dots\dots\dots$$

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 5****TOPIC: Vectors**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use vectors**

$$\overrightarrow{OA} = 4a, \quad \overrightarrow{OB} = 6b, \quad \overrightarrow{AP} = 2a + 8b$$

2. Use vectors

So

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$$

$$\overrightarrow{OP} = 4a + (2a + 8b) = 6a + 8b$$

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Substitute:

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$$4 - 4\mu = \frac{9}{2}\mu$$

$$8 = 17\mu$$

$$\mu = \frac{8}{17}$$

10. Use vectors

Therefore

$$AQ : QB = \frac{8}{17} : \frac{9}{17} = 8 : 9$$

FINAL ANSWER

$$AQ : QB = 8 : 9.$$

PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Sequences**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 24** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.
The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 5****TOPIC: Sequences**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the first term be**Let the first term be a and the common difference be d .

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

2. Find the nth term

So

$$S_{10} = 5(2a + 9d)$$

3. Find the nth term

and

$$S_5 = \frac{5}{2}(2a + 4d)$$

4. GivenGiven $S_{10} = 4S_5$:

$$5(2a + 9d) = 4\left(\frac{5}{2}(2a + 4d)\right)$$

$$10a + 45d = 20a + 40d$$

$$5d = 10a$$

$$d = 2a$$

5. The 8th term is

The 8th term is 45:

$$a + 7d = 45$$

6. SubstituteSubstitute $d = 2a$:

$$a + 14a = 45$$

$$15a = 45$$

$$a = 3$$

FINAL ANSWER

the first term is 3.

EA

PAST PAPER QUESTION

Q#: 24

Marks: 5

TOPIC: **Sequences**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 24** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.
The 8th term of this series is 45
Find the first term of this series.
Show clear algebraic working.

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 5****TOPIC: Sequences**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the first term be**Let the first term be a and the common difference be d .

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$$d = 2a$$

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The 8th term is 45:

$$a + 7d = 45$$

6. SubstituteSubstitute $d = 2a$:

$$a + 14a = 45$$

$$15a = 45$$

$$a = 3$$

FINAL ANSWER

the first term is 3.

EA